

Reference Layer Models

What are Reference Layer Models?

A reference layer model represents the complex communication within networks, structured and divided into layers. Each layer has clearly defined tasks and offers a standardized interface. Additionally, by using the same reference model, different companies can develop products that are compatible with each other.

Nowadays, with very rare exceptions, three different layer models are used: OSI, TCP/IP and IEEE 802.

Purposes of Reference Layer Models

- Reduction of complexity
- Modularization
- Interoperability across different manufacturers
- Easier fault localization
- Independence from specific hardware and technologies

The OSI Model

OSI stands for Open Systems Interconnection. It was initiated by the ISO (International Organization for Standardization) in 1977 and was published as ISO 7498 in 1984.

The OSI model was developed as a universal standard model for all networks.

Layers and Tasks of the OSI Model

7	Application Layer	Interface to applications (e.g., HTTP, SMTP)
6	Presentation Layer	Data formatting (e.g., ASCII, JPEG, encryption, compression)
5	Session Layer	Session management, dialog control (e.g., session resumption)
4	Transport Layer	End-to-end communication, error correction, flow control (TCP, UDP)
3	Network Layer	Routing, logical addressing (e.g., IP, ICMP, ARP)
2	Data Link Layer	Error control, framing, access control (e.g., MAC addresses)
1	Physical Layer	Transmission of bits via plugs, cables, and/or signal levels

The OSI Model in Practice

The OSI model is used for structuring protocols, network theory, and teaching. Practical implementations only loosely follow it.

The TCP/IP Model

ARPA, later DARPA ((Defense) Advanced Research Projects Agency), developed the TCP/IP model in the 1970s for its ARPANET (Advanced Research Projects Agency NETwork). In 1983, TCP/IP became the standard protocol in ARPANET.

Interesting fact: ARPANET was designed to enable infrastructure destruction during wartime while being able to survive even a nuclear attack itself. It is the direct predecessor of today's internet. ARPA was founded in 1958, one year after the "Sputnik shock", as the USA wanted to avoid technological surprises in the future. The "Defense" addition (ARPA → DARPA) occurred in 1972.

Layers and Tasks of the TCP/IP Model

5	Application Layer	All application-oriented protocols (e.g., HTTP, FTP, SMTP) (OSI 5-7)
4	Transport Layer	End-to-end communication (TCP, UDP)
3	Network Layer	Routing and logical addressing (e.g., IP, ICMP, ARP)
2	Data Link Layer	Error control, framing, access control (e.g., MAC addresses)
1	Physical Layer	Transmission via plugs, cables, and/or signal levels

Note: Originally, the TCP/IP model had only four layers, with the first two combined into a so called Network Access Layer. The remaining layers are then numbered one lower. However, the 5-layer TCP/IP model is more practical and functionally accurate than the 4-layer version.

The TCP/IP Model in Practice

The TCP/IP model is the practical implementation of the Internet. It is more flexible than the OSI model and specifically tailored to the internet protocol family ([IPv4](#), [IPv6](#), HTTP, FTP, DNS, DHCP, SMTP, POP3, etc.). Thus, it is used in network engineering and administration as a practical layer model.

The IEEE 802 Model

The Institute of Electrical and Electronics Engineers (IEEE) developed the IEEE 802 model in 1980, which has since specified standards for local area networks (LANs) and metropolitan area networks (MANs). This reference model is limited to the first two layers of the OSI model or the 5-layer TCP/IP model.

Note: "802" is simply the project number. The IEEE established a committee for the standardization of LANs and MANs in February (the 2nd month) of 1980, and the project number was formed by the year (80) and the month (2).

Layers and Tasks of the IEEE 802 Model

1. **Physical Layer:** Transmission of bits via plugs, cables, and/or signal levels.
2. **Data Link Layer:** Further divided into:
 - a) **Logical Link Control (LLC, IEEE 802.2):** Frame control, error handling, flow control.
 - b) **Media Access Control (MAC, IEEE 802.3):** Access control, addressing, collision avoidance.

Interesting: With over 400,000 members, the IEEE is the world's largest professional organization for engineers. It was formed in 1963 from the merger of the AIEE (American Institute of Electrical Engineers, founded in 1884) and the IRE (Institute of Radio Engineers, founded in 1912).

Significance

IEEE 802 plays a central role in the development of physical networks and LAN technologies. For hardware manufacturers and network protocols, these standards are essential as they create a unified basis for compatibility and interoperability, ensuring that devices from different manufacturers can work together seamlessly.

Examples

IEEE 802 includes numerous sub-standards defining various network technologies:

- IEEE 802.3 (Ethernet): The best-known standard for wired LANs.
- IEEE 802.11 (WLAN): Specifies wireless network technologies (Wi-Fi).
- IEEE 802.15 (Bluetooth/Zigbee): Defines personal area networks (PANs) for Bluetooth and Zigbee communication.

Comparison of the Models

Aspect	OSI	TCP/IP	IEEE 802
Layers	7	5 (4)	2
Focus	Universal teaching model	Practical internet model	LAN/MAN, Layers 1 and 2
Usage	Teaching, structuring	Internet, real networks	LAN standards (Ethernet, WLAN)
Modularity	High	Less high	Limited to layers 1 and 2
Flexibility	High, but complex	Very flexible and pragmatic	Optimized for physical networks

Interesting Details and Trivia

- OSI and TCP/IP were developed in parallel, but TCP/IP gained practical dominance through ARPANET/Internet, while OSI remained confined to academia.
- The OSI model also defines services (connection-oriented and connectionless) and primitive operations, which are not explicitly specified in TCP/IP.
- IEEE 802 standards form the physical foundation of the internet (Ethernet, WLAN), while TCP/IP operates on top.
- OSI layer 5 (Session) and 6 (Presentation) are rarely implemented separately on the internet, with their functions mostly integrated into the Application Layer.
- Modern protocols like QUIC (over UDP) illustrate how flexible layer usage has become.
- Modern networks often refer to a layered protocol stack, but with hybrid implementation (e.g., VPNs (Layer 2/3, VLANs (Layer 2), or TLS (Layer 6 over Layer 4).

Other (Historical) Reference Layer Models

SNA

System Network Architecture (SNA) was developed by IBM in 1974 as a hierarchical network architecture for mainframe communication, defining clear communication layers but exclusive for IBM systems, never as a open standard. With its seven layers, it roughly resembles the OSI model. The layers were:

7	Transaction Services Layer	Application logic
6	Presentation Services Layer	Data formatting
5	Data Flow Control Layer	Session control
4	Transmission Control Layer	Connection management
3	Path Control Layer	Routing, packet switching
2	Data Link Control Layer	Framing, error control
1	Physical Control Layer	Bit transmission

SNA was simplified in the 1980s into APPN and APPC, still occasionally used in legacy systems like banks, but always remained IBM-specific.

DECnet

Decnet was developed by the Digital Equipment Corporation (DEC) in 1975, based on a 5-layer model. The layers of the most common version (Phase IV) were:

5	Application Layer	File transfer services
4	Network Layer	Routing, addressing
3	Transport Layer	End-to-end communication
2	Data Link Layer	MAC addresses, error detection
1	Physical Layer	Hardware e.g., plugs, cables, ethernet

DECnet Phase V migrated to OSI standards in 1987 to maintain compatibility with international technology trends. At that time, both OSI and TCP/IP were emerging as key standards in the IT world, though TCP/IP ultimately won out.

DEC was acquired by Compaq in 1998.

AppleTalk

In 1985, Apple introduced AppleTalk as a protocol suite for local networks. Although AppleTalk included its own layer concepts, these were never formally standardized and remained limited to Apple devices. It used an informal, 6-layer model.

6	Application Layer	e.g., Apple Filing Protocol (AFP)
5	Session Layer	AppleTalk Session Protocol (ASP)
4	Transport Layer	AppleTalk Transaction Protocol (ATP)
3	Network Layer	Datagram Delivery Protocol (DDP)
2	Data Link Layer	LocalTalk Link Access Protocol (LLAP)
1	Physical Layer	e.g., LocalTalk via RS-442

AppleTalk was completely removed from Apple operating systems between 1999 and 2009 and replaced by TCP/IP-compatible protocols.

X.25

An early standard for packet-switched networks was X.25, developed by ITU-T in 1976. It dominated before TCP/IP, particularly in Europe, and served as the foundation for many early commercial data networks (e.g., Datex-P in Germany). It was a 3-layer model:

3	Network Layer (PLP)	Addressing via X.121 addresses (a precursor to IP addresses)
2	Link Layer (LAP-B)	Error correction per hop ("reliable wires")
1	Physical Layer	RS-232, V.24 (analog) or ISDN (digital)

Fun fact: TCP/IP adopted X.25's error correction principle but shifted it from hop-by-hop (each network node verifies) to end-to-end control. This reduced routers to "dumb" packet forwarding, boosting the internet's speed and scalability.

Ironically, modern mobile networks like LTE/5G revived the per-hop error correction, because wireless links are more error-prone.

Even today, some banks still use X.25 for ATM transactions. It also remains in limited use in aviation (ACARS) and railway signaling systems.

Note: The International Telecommunications Union - Telecommunication Standardization Sector (ITU-T) is the "technical arm" of the ITU. Originally founded as the International Telegraph Union on May 17, 1865, in Paris. In 1932 the "Telegraph" was replaced by "Telecommunication".

